



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-23

Experiment 23. Prolog Program to Suggest Dieting System Based on Disease

Aim

To write a Prolog program to suggest a diet based on a disease.

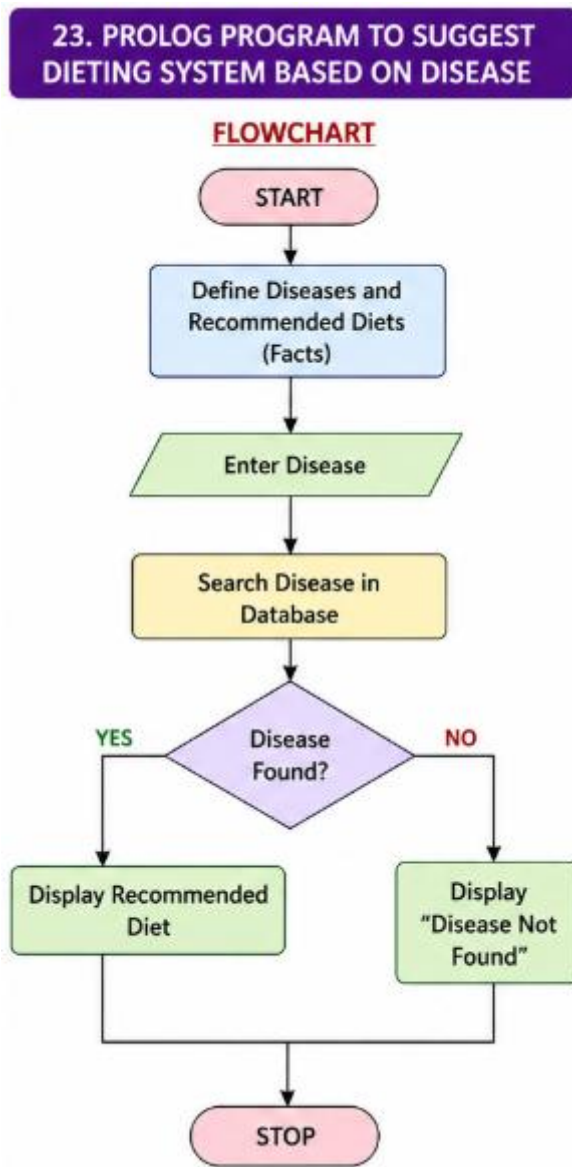
Objectives

- To understand facts and rules in Prolog.
- To associate diseases with suitable diets.
- To generate a diet suggestion.

Algorithm

1. Define diseases and their recommended diets.
2. Select a disease.
3. Search the database.
4. Display the recommended diet.

Flowchart



Code

```
diet(diabetes, "Low sugar  
diet").
```

```
diet(obesity, "Low calorie  
diet").
```

```
diet(anemia, "Iron rich  
diet").
```

```
diet(hypertension, "Low  
salt diet").
```

main :-

```
    Disease = diabetes,  
  
    diet(Disease, Diet),  
  
    write('Disease: '),  
write(Disease), nl,  
  
    write('Recommended  
Diet: '), write(Diet), nl.  
  
:- initialization(main).
```

Output

```

HelloWorld.pl x
1  diet(diabetes, "Low sugar diet").
2  diet(obesity, "Low calorie diet").
3  diet(anemia, "Iron rich diet").
4  diet(hypertension, "Low salt diet").
5
6  main :-
7      Disease = diabetes,
8      diet(Disease, Diet),
9      write('Disease: '), write(Disease), nl,
10     write('Recommended Diet: '), write(Diet), nl.
11
12     :- initialization(main).
```

I/O AI Agent 121 ms

STDIN

Input for the program (Optional)

Output

Disease: diabetes

Recommended Diet: [76,111,119,32,115,117,103,97,114,32,100,105,101,116]

Result

Disease: diabetes

Recommended Diet: Low sugar diet

Conclusion

Thus, the Prolog program successfully suggests a suitable diet based on the given disease.